EVALUATION REPORT

AI-Powered Energy Consumption Optimization Assistant

Implemented evaluation methods: DeepEval, custom LLM-as-Judge, retrieval metrics, risk scoring, and API/workflow validation.

Evaluation framework	DeepEval (RAGAS not implemented)
Additional evaluation	Custom LLM-as-Judge and retrieval metrics
Application	FastAPI backend and Streamlit UI, run locally

Scores are observations from specific test runs, not guarantees of general performance.

1. Executive Summary

This report documents the evaluation capabilities implemented in the Energy Consumption Optimization Assistant. The system combines semantic retrieval with keyword search, query classification, recommendation generation, risk scoring, and a Streamlit interface backed by FastAPI.

DeepEval evaluates answer relevance and energy recommendation quality. A separate custom LLM-as-Judge service validates recommendations. Retrieval quality is assessed with Precision, Recall, and F1 for selected labelled queries. API and end-to-end workflow checks provide functional validation.

> RAGAS is intentionally excluded because it is not implemented in this project.

2. Implemented Evaluation Methods

2.1 DeepEval

DeepEval uses an evaluation LLM configured separately through `DEEPEVAL_MODEL`. Implemented metrics include Answer Relevancy and a GEval metric named *Energy Recommendation Quality*. The service records metric scores, success status, reasons, and an overall score based on available metric scores.

2.2 Custom LLM-as-Judge

A separate service evaluates the query, context, and recommendation using an LLM and parses the result into a structured response. It is configured independently through `LLM_JUDGE_MODEL`. A recent standalone test returned success with an overall score of 0.70.

2.3 Retrieval metrics

Precision, Recall, and F1 are used for selected test queries where expected relevant cases are available. These metrics assess retrieval against the labels used; they do not directly measure recommendation correctness.

2.4 Risk prediction

The risk component returns a demand-risk category and score for an input. Because anomaly/risk labels are derived from processing logic and have not been independently validated against real operational incidents, these are experimental outputs—not proven real-world risk probabilities or independently verified accuracy.

2.5 API and workflow checks

During local validation, the health, recommendation, and evaluation endpoints returned HTTP 200 OK, and the Streamlit interface displayed results. HTTP 200 confirms the endpoint returned successfully; individual metric success fields must also be checked.

3. Observed DeepEval Results

These scores come from specific test runs shared during development. They are example-run results, not an aggregate benchmark across a large test set.

Test scenario	Mode	DeepEval overall	Risk output	Cases
High consumption during peak hours	LLM + RAG	89.63%	MEDIUM (98.99% model score)	5
Similar high usage under similar conditions	LLM + RAG	95.00%	NOT_ASSESSED	5
High consumption despite prior optimization	LLM + RAG	98.18%	HIGH (87.84% model score)	5
High consumption during peak hours	Rule-based	28.18%	Workflow-dependent	5
Similar high usage under similar conditions	Rule-based	88.93%	NOT_ASSESSED	5
High consumption despite prior optimization	Rule-based	90.22%	Workflow-dependent	5

Recent standalone evaluator checks

Evaluation	Observed result	Interpretation
DeepEval	Answer Relevancy 0.75; Energy Recommendation Quality 0.90; overall 0.825 (82.5%)	Both metrics returned scores in this run.

Evaluation	Observed result	Interpretation
Custom LLM-as-Judge	Success: true; overall score 0.70	Judge returned a score in this run.

4. Retrieval Evaluation Results

The following values were reported for selected labelled queries. They are not universal system-wide results.

Query scenario	Precision	Recall	Reported F1
High consumption during peak hours	100%	0.42%	0.83%
High consumption despite prior optimization	100%	0.22%	0.44%

Interpretation and data-quality note: Very low recall suggests many expected relevant cases were not retrieved, or that the relevance labels/test denominator need review. The recorded F1 values do not appear mathematically consistent with the displayed precision and recall under the standard F1 formula. Recalculate them from the evaluation code before using them as formal benchmark results.

5. Functional and API Validation

Observed local checks: FastAPI health endpoint returned 200 OK; `POST /api/recommend/llm` returned 200 OK; `POST /api/evaluate` returned 200 OK; Streamlit displayed recommendations and evaluation results; standalone LLM-as-Judge and DeepEval tests returned scores.

6. Limitations and Threats to Validity

Dataset scope: The Appliances Energy Prediction dataset represents one household/building environment, not multiple independent facilities. Multi-facility questions are answered using analogous historical patterns, not actual multi-facility records.

Derived labels: Risk/anomaly labels are derived from processing logic and are not independently validated against operational incidents. Do not claim proven real-world accuracy or calibrated risk probabilities.

Small evaluation sample: The DeepEval scores shown come from a small number of queries and separate runs, not a statistically robust benchmark.

Retrieval recall: Recall is low for the selected labelled queries and should be investigated.

DeepEval schema warning: A recent run warned that the provider rejected the strict structured-output schema because the required list omitted the 'reason' property. DeepEval attempted regular generation with JSON parsing and still returned scores in that run. This compatibility warning does not guarantee all future evaluations will succeed.

Cost/carbon estimates: Treat savings as estimates unless verified with documented tariff rates, emissions factors, and measured before/after consumption.

LLM judge: Automated judge scores are model-based assessments, not ground-truth labels; they may vary by model, prompt, context, and provider.

7. Recommended Next Steps

- Create a fixed, manually reviewed set of energy queries, relevant cases, and reference recommendations.
- Recalculate Precision, Recall, and F1 from documented labels and verify the F1 implementation.
- Run DeepEval over the same fixed test set and report sample size, metric averages, and failed evaluations.
- Keep custom LLM-as-Judge results separate from DeepEval unless a combined score has a defined weighting method.
- Validate risk outputs against independent labels before reporting accuracy or calibrated probabilities.
- Document assumptions and sources for cost/carbon estimates.

8. Conclusion

The project includes evaluation components for recommendation quality (DeepEval), automated recommendation validation (custom LLM-as-Judge), retrieval metrics, risk scoring, and API/UI checks. Reported runs show these components can return results, while highlighting improvement areas: retrieval recall, small evaluation sample, and a structured-output compatibility warning. RAGAS is excluded because it is not implemented.

Report note: Values are transcribed from project test outputs shared during development. Re-run tests before submission if code, prompts, data, or model configuration changes.